Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 1 of 10 · Question 6 part a.i · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

6(a)(i)	$\frac{dy}{dx} = 4e^{2x} + 2(4x-3)e^{2x}$	M1A1A1 (3)
(ii)	$(4x-3)\frac{dy}{dx} = (4x-3)(8x-2)e^{2x} = (8x-2)y$ *	M1A1cso (2)
(b)	$\frac{dy}{dx} = \frac{5\cos 5x \times (x-3)^2 - \sin 5x \times 2(x-3)}{(x-3)^4}$	M1A1A1 (3)
ALT	Using product rule:	
	$y = (x-3)^{-2}\sin 5x$	M1
	$\frac{dy}{dx} = -2(x-3)^{-3}\sin 5x + 5(x-3)^{-2}\cos 5x$	A1A1
		[8]

(a)(i)	
M1	Use product rule to differentiate the given expression. Must have 2 terms added. One to be of the form ke^{2x} and the other of the form $k'(4x-3)e^{2x}$ where k' = 1 or 2
A1	Either term correct
A1	Second term correct
NB	No simplification needed for these 3 marks
ALT	$y = 4xe^{2x} - 3e^{2x} \Rightarrow \frac{dy}{dx} = 4e^{2x} + 8xe^{2x} - 6e^{2x}$
	M1 Expand the given expression and differentiate using the product rule for $4xe^{2x}$ A1 Any 2 terms correct; A1 Third term correct.
NB	No simplification needed for these 3 marks
(ii)	
M1	Use their result from (i) to obtain an expression for $(4x-3)\frac{dy}{dx}$. No need to simplify.
A1cso	Correct **given** result obtained with no errors in the working. Can start with LHS and show equal to the RHS or vice versa or can start with each side and "meet in the middle"
(b)	
M1	Attempt the quotient rule. The denominator must be $(x-3)^4$ and the numerator must be of the form $\left(k\cos 5x \times (x-3)^2 - \sin 5x \times l(x-3)\right)$ $k = \pm 5$ or ± 1, $l = 1$ or 2 (ie sine may have been differentiated to − cosine)
A1	One fully correct term in numerator.
A1	All fully correct.
ALT	
M1	Rewrite without a quotient and apply the product rule obtaining 2 terms of the form shown
A1	Either term correct
A1	Second term correct No need to simplify

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 1 of 10 · Question 6 part a.i · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 2 of 10 · Question 1 part whole · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

1	$36xe^{3x^2}\cos 2x - 12e^{3x^2}\sin 2x$	M1A1A1 (3) [3]

Mark	Notes
	$6e^{3x^2}\cos 2x$
M1	For applying the Product rule • There must be an attempt to differentiate both terms. Accept as a minimum **either** $e^{3x^2} \Rightarrow \pm axe^{3x^2}$ **or** $\cos 2x \Rightarrow -b\sin 2x$ • A correct application of product rule – accept e.g $36xe^{3x^2}\cos 2x \pm 12e^{3x^2}\sin 2x$ $\left[36xe^{3x^2}\cos 2x - 12e^{3x^2}\sin 2x\right]$
A1	For either $36xe^{3x^2}\cos 2x$ or $-12e^{3x^2}\sin 2x$ Need not be simplified
A1	For the fully correct expression $36xe^{3x^2}\cos 2x - 12e^{3x^2}\sin 2x$ Need not be simplified. Accept for example: $6\times 6xe^{3x^2}\cos 2x - 6\times 2\times e^{3x^2}\sin 2x$

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 2 of 10 · Question 1 part whole · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 3 of 10 · Question 11 part a · 5 marks

2022 JAN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

11 (a)	$\frac{e^{4x}}{32}$ **and** $8x^2-4x+1$	M1
	$\frac{dy}{dx}=\frac{e^{4x}}{32}(16x-4)+\frac{e^{4x}}{8}(8x^2-4x+1)$	A1A1
	$\frac{dy}{dx}=\frac{16xe^{4x}}{32}-\frac{4e^{4x}}{32}+\frac{8x^2e^{4x}}{8}-\frac{4xe^{4x}}{8}+\frac{e^{4x}}{8}$	dM1
	$\frac{dy}{dx}=x^2e^{4x}$ *	A1 cso [5]
(b)	Volume $=\pi\int_{-2}^{0}(3xe^{2x})^2\,dx$	M1
	$V=\pi\int_{-2}^{0}9x^2e^{4x}\,dx=\pi\left[\frac{9e^{4x}}{32}(8x^2-4x+1)\right]_{-2}^{0}$	M1
	$\pi\int_{-2}^{0}9x^2e^{4x}\,dx=\pi\left[\frac{9e^{4\times0}}{32}(8\times0^2-4\times0+1)\right]-\pi\left[\frac{9e^{4\times-2}}{32}(8\times-2^2-4\times-2+1)\right]$	dM1
	$V=0.87142.....\approx0.87$ (2sf)	A1 [4]
	SC - attempts integration by parts.	
	Volume $=\pi\int_{-2}^{0}(3xe^{2x})^2\,dx$	M1
	$\int x^2e^{4x}\,dx=\frac{x^2e^{4x}}{'4'}-\int2x\frac{e^{4x}}{'4'}\,dx$ $\int x^2e^{4x}\,dx=\frac{x^2e^{4x}}{'4'}-\left[\frac{2xe^{4x}}{'16'}-\int\frac{2e^{4x}}{'16'}\right]=\frac{x^2e^{4x}}{'4'}\pm\frac{2xe^{4x}}{'16'}\pm\frac{e^{4x}}{'32'}$	M1
	$\pi\int_{-2}^{0}9x^2e^{4x}\,dx=9\pi\left[\frac{x^2e^{4x}}{'4'}-\frac{2xe^{4x}}{'16'}+\frac{e^{4x}}{'32'}\right]_{-2}^{0}$ $=9\pi\left[\left(0-0+\frac{1}{32}\right)-\left(\frac{(-2)^2e^{4(-2)}}{'4'}-\frac{2(-2)e^{4(-2)}}{'16'}+\frac{e^{4(-2)}}{'32'}\right)\right]$	dM1
	For the correct volume of $0.87142.....\approx0.87$ rounded correctly to 2sf	A1 [4]
		Total 9 marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 3 of 10 · Question 11 part a · 5 marks

2022 JAN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using product rule correctly with an attempt to differentiate both $\frac{e^{4x}}{32}$ **and** $8x^2-4x+1$ Correct application of $\frac{dy}{dx}=u\frac{dv}{dx}+v\frac{du}{dx}$ is required For attempt to differentiate: $8x^2-4x+1\rightarrow 16x-4$ $\frac{e^{4x}}{32}\rightarrow ke^{4x}$ where $k\neq 1$ or $\frac{1}{32}$ or $\frac{1}{128}$
	A1	For either $\frac{e^{4x}}{32}(16x-4)$ OR $\frac{e^{4x}}{8}(8x^2-4x+1)$ o.e. in both cases
	A1	For $\frac{dy}{dx}=\frac{e^{4x}}{32}(16x-4)+\frac{e^{4x}}{8}(8x^2-4x+1)$ fully correct Accept $\frac{dy}{dx}=\frac{e^{4x}}{32}(16x-4)+\frac{4e^{4x}}{32}(8x^2-4x+1)$
	dM1	For multiplying out both sets of brackets, or factorising, their $\frac{dy}{dx}$ provided the earlier M mark has been achieved e.g., $\frac{dy}{dx}=\frac{16xe^{4x}}{32}-\frac{4e^{4x}}{32}+\frac{8x^2e^{4x}}{8}-\frac{4xe^{4x}}{8}+\frac{e^{4x}}{8}$ Some working to collect terms between product rule and final expression.
	A1 cso	For the correct expression only $\frac{dy}{dx}=x^2e^{4x}$ *
(b)	M1	This is a strategy mark for using the correct expression with the correct limits for the volume of rotation. Volume $=\pi\int_{-2}^{0}\left(3xe^{2x}\right)^2\,dx$
	M1	For using the **given** result from part (a) to integrate $V=\pi\int_{-2}^{0}9x^2e^{4x}\,dx=\pi\left[\frac{9e^{4x}}{32}(8x^2-4x+1)\right]_{-2}^{0}$ Ignore π and limits for this mark even if they are missing or incorrect.
	dM1	For applying the correct limits in an attempt to evaluate the integral $\pi\int_{-2}^{0}9x^2e^{4x}\,dx=\pi\left[\frac{9e^{4\times0}}{32}(8\times0^2-4\times0+1)\right]-\pi\left[\frac{9e^{4\times-2}}{32}(8\times-2^2-4\times-2+1)\right]$ Dependent on previous method mark. Condone omission of π for this mark.
	A1	For the correct volume of $0.87142\ldots\approx 0.87$ rounded correctly to 2sf
SC -attempts integration by parts.		
	M1	This is a strategy mark for using the correct expression with the correct limits for the volume of rotation.

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 3 of 10 · Question 11 part a · 5 marks

2022 JAN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

		Volume $= \pi\int_{-2}^{0}\left(3xe^{2x}\right)^2\ dx$
	M1	For an attempt to integrate by parts. • They must use the correct formula • $\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$ **TWICE** in the correct direction First integration – ignore π and 3^2 for this mark $\int x^2e^{4x}\,dx = \frac{x^2e^{4x}}{'4'} - \int 2x\frac{e^{4x}}{'4'}\,dx$ Second integration $\int x^2e^{4x}\,dx = \frac{x^2e^{4x}}{'4'} - \left[\frac{2xe^{4x}}{'16'} - \int\frac{2e^{4x}}{'16'}\right] = \frac{x^2e^{4x}}{'4'} \pm \frac{2xe^{4x}}{'16'} \pm \frac{e^{4x}}{'32'}$
	dM1	For applying the correct limits in an attempt to evaluate the integral $\pi\int_{-2}^{0}9x^2e^{4x}\,dx = 9\pi\left[\frac{x^2e^{4x}}{'4'} - \frac{2xe^{4x}}{'16'} + \frac{e^{4x}}{'32'}\right]_{-2}^{0}$ $= 9\pi\left[\left(0-0+\frac{1}{32}\right) - \left(\frac{(-2)^2e^{4(-2)}}{'4'} - \frac{2(-2)e^{4(-2)}}{'16'} + \frac{e^{4(-2)}}{'32'}\right)\right]$ Dependent on previous method mark. Condone omission of π for this mark.
	A1	For the correct volume of $0.87142..... \approx 0.87$ rounded correctly to 2sf

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 3 of 10 · Question 11 part a · 5 marks

2022 JAN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

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Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 4 of 10 · Question 3 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

3 (a)	$\frac{dy}{dx} = 2e^{2x}\sqrt{5x-3} + \frac{5e^{2x}}{2\sqrt{5x-3}}$	M1 A1 A1 (3)
(b)	$\frac{dy}{dx} = \frac{3x^2\cos 3x - x^3(-3\sin 3x)}{(\cos 3x)^2}$	M1 A1 A1 (3)
		Total 6 marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 4 of 10 · Question 3 part a · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 5 of 10 · Question 6 part b · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

6(a)(i)	$\frac{dy}{dx} = 4e^{2x} + 2(4x-3)e^{2x}$	M1A1A1 (3)
(ii)	$(4x-3)\frac{dy}{dx} = (4x-3)(8x-2)e^{2x} = (8x-2)y$ *	M1A1cso (2)
(b)	$\frac{dy}{dx} = \frac{5\cos 5x \times (x-3)^2 - \sin 5x \times 2(x-3)}{(x-3)^4}$	M1A1A1 (3)
ALT	Using product rule:	
	$y = (x-3)^{-2}\sin 5x$	M1
	$\frac{dy}{dx} = -2(x-3)^{-3}\sin 5x + 5(x-3)^{-2}\cos 5x$	A1A1
		[8]

(a)(i)	
M1	Use product rule to differentiate the given expression. Must have 2 terms added. One to be of the form ke^{2x} and the other of the form $k'(4x-3)e^{2x}$ where k' = 1 or 2
A1	Either term correct
A1	Second term correct
NB	No simplification needed for these 3 marks
ALT	$y = 4xe^{2x} - 3e^{2x} \Rightarrow \frac{dy}{dx} = 4e^{2x} + 8xe^{2x} - 6e^{2x}$
	M1 Expand the given expression and differentiate using the product rule for $4xe^{2x}$ A1 Any 2 terms correct; A1 Third term correct.
NB	No simplification needed for these 3 marks
(ii)	
M1	Use their result from (i) to obtain an expression for $(4x-3)\frac{dy}{dx}$. No need to simplify.
A1cso	Correct **given** result obtained with no errors in the working. Can start with LHS and show equal to the RHS or vice versa or can start with each side and "meet in the middle"
(b)	
M1	Attempt the quotient rule. The denominator must be $(x-3)^4$ and the numerator must be of the form $\left(k\cos 5x \times (x-3)^2 - \sin 5x \times l(x-3)\right)$ $k = \pm 5$ or ± 1, $l = 1$ or 2 (ie sine may have been differentiated to − cosine)
A1	One fully correct term in numerator.
A1	All fully correct.
ALT	
M1	Rewrite without a quotient and apply the product rule obtaining 2 terms of the form shown
A1	Either term correct
A1	Second term correct No need to simplify

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 5 of 10 · Question 6 part b · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 6 of 10 · Question 7 part a · 4 marks

2023 JAN · Paper 02 · find part a in this mark scheme extract

7 a	$\frac{dy}{dx} = 2e^{2x}\cos 2x - 2e^{2x}\sin 2x$	M1 A1 A1
	$\frac{dy}{dx} = 2y - 2e^{2x}\sin 2x$ *	A1 cso (4)
b	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$	M1 A1
	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2\left(\frac{dy}{dx} - 2y\right) - 4y$	dM1ddM1
	$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$ *	A1 cso (5)
		Total 9 marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 7 of 10 · Question 9 part a · 3 marks

2025 JUN · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

9 (a)	$\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ *	M1A1A1 Cso [3]
(b)	**Method A** $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$ $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y \Rightarrow \frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$ $\Rightarrow \frac{d^2y}{dt^2} = -8\frac{dy}{dt} - 20y \Rightarrow \frac{d^2y}{dt^2} + 8\frac{dy}{dt} + 20y = 0$ $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0 \Rightarrow M = 8, N = 20$	M1 M1 M1 M1 A1 [5]
	Method B $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ $\left(= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t\right)$ $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$ $= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$ $= (16 - 2M)e^{-4t}\sin 2t + (12 - 4M + N)e^{-4t}\cos 2t$ $16 - 2M = 0$ and $12 - 4M + N = 0$ $M = 8,\ N = 20$	[M1 M1 M1 M1 A1]
	Method C By inspection – please see additional guidance.	
		Total 8 marks

Part	Mark	Additional Guidance

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 7 of 10 · Question 9 part a · 3 marks

2025 JUN · Paper 02 · extract 2 of 4 · find part a in this mark scheme extract

(a)	M1	For an attempt to differentiate the given expression. • There needs to be an acceptable attempt to differentiate both terms. $e^{-4t}\cos 2t \rightarrow ke^{-4t}\sin 2t + le^{-4t}\cos 2t$ with $k, l \neq 0$ • There need to be two terms added. $\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t$
	A1	One term completely correct. $\frac{dy}{dt} = -4e^{-4t}\cos 2t + '-2e^{-4t}\sin 2t$ OR $\frac{dy}{dt} = '-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t$
	A1	Fully correct differentiated expression and rearranged into the required form $\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ Accept also, For example: $-\left(-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t\right) - 4\left(e^{-4t}\sin 2t\right) = 2e^{-4t}\sin 2t$ With any unambiguous indication that a conclusion has been reached, e.g., QED, hash sign, double underline, RHS = LHS etc.
		Method A
(b)	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4x}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For substituting y and $\frac{dy}{dt}$ into their $\frac{d^2y}{dt^2}$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$
	M1	For preparing to eliminate $\cos 2t$ by rearranging their $\frac{dy}{dt}$ $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$ Allow errors in arithmetic but not mathematically incorrect process.
	M1	For an unsimplified expression only in terms of y, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$
	A1 [8]	For finding the value of M and the value of N. $M = 8,\ N = 20$
		Method B

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 7 of 10 · Question 9 part a · 3 marks

2025 JUN · Paper 02 · extract 3 of 4 · find part a in this mark scheme extract

	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4t}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For substituting y, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ into the equation. $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$ $= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$
	M1	For collecting terms $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = ('16' - 2M)e^{-4t}\sin 2t + ('12' - 4M + N)e^{-4t}\cos 2t$
	M1	For identifying two equations from their working and a method to solve them simultaneously For example, $'16' - 2M = 0$ and $'12' - 4M + N = 0$
	A1	For finding the value of M and the value of N. $M = 8, N = 20$
		Method C (By inspection)
	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4t}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For simplifying $\frac{d^2y}{dt^2} = e^{-4t}16\sin 2t + e^{-4t}12\cos 2t = \left[e^{-4t}(16\sin 2t + 12\cos 2t)\right]$
	M1	For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}[\sin 2t]$ and deducing that $'16' + (\underline{8} \times -2) = 0 \Rightarrow M = 8$
	M1	For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}$ and y and deducing that $'12' - ('8' \times 4) + \underline{20} = 0 \Rightarrow N = 20$
	A1	For both $M = 8$ and $N = 20$
		Total 8 marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 7 of 10 · Question 9 part a · 3 marks

2025 JUN · Paper 02 · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 8 of 10 · Question 3 part a · 3 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

3(a)	$\frac{dy}{dx} = 2e^{2x}\left(x^2+1\right)+e^{2x}\left(2x\right)$	M1A1A1 [3]
(b)	When $x = 0$ $\frac{dy}{dx} = 2\times1\times1+1\times0 = 2$ $\quad y = e^{2\times0}\left(0+1\right) = 1$ $y-1 = 2\left(x-0\right) \Rightarrow y = 2x+1$	B1B1 B1 [3]
		Total 6 marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 8 of 10 · Question 3 part a · 3 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks
2 (a)	$7+4x-x^2=11-(x-2)^2$ $[a=11, b=1, c=-2]$ **ALT**	M1A1A1 [3]

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 9 of 10 · Question 5 part a · 5 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

5 (a)	$4e^{4x}(6x+2)^{\frac{3}{2}}+\frac{3}{2}.6.e^{4x}(6x+2)^{\frac{1}{2}}=\left[4e^{4x}(6x+2)^{\frac{3}{2}}+9e^{4x}(6x+2)^{\frac{1}{2}}\right]$ $e^{4x}\sqrt{6x+2}(24x+17)$	M1 A1 A1 dM1 A1 [5]
(b)	$\frac{3\cos 3x(2x-4)^3-\sin 3x.2.3.(2x-4)^2}{\left[(2x-4)^3\right]^2}=\left[\frac{3\cos 3x(2x-4)^3-6\sin 3x(2x-4)^2}{(2x-4)^6}\right]$ OR $\frac{3\cos 3x(2x-4)^3-\left(24x^2-96x+96\right)\sin 3x}{(2x-4)^6}$ ALT using product rule $\sin 3x(2x-4)^{-3}$ $3\cos 3x(2x-4)^{-3}+\sin 3x(-3)(2)(2x-4)^{-4}=\left[3\cos 3x(2x-4)^{-3}-6\sin 3x(2x-4)^{-4}\right]$	M1 A1 A1 M1 A1 A1 [3]
	Total	**8 marks**

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 9 of 10 · Question 5 part a · 5 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Use of product rule to give an expression of the form $pe^{4x}(6x+2)^{\frac{3}{2}}+qe^{4x}(6x+2)^{\frac{1}{2}}$ where $p = 1$ or 4, $q > 1$. There must be a + sign between terms.
	A1	Either term correct. Note that simplification is not required for this mark.
	A1	Both terms correct. Note that simplification is not required for this mark.
	dM1	Obtains an expression of the form $e^{4x}\left(\sqrt{6x+2}\right)(Ax+B)$ where $A, B \neq 0$ You **must** check their working here that it follows from their attempt at product rule. There must be an intermediate step seen as follows for example: $4e^{4x}(6x+2)^{\frac{3}{2}}+9e^{4x}(6x+2)^{\frac{1}{2}}=e^{4x}(6x+2)^{\frac{1}{2}}\left(4\left(6x+2\right)+9\right)$ **Note: This mark is dependent on the first M Mark**
	A1	Correct $A = 24$ and correct $B = 17$
(b)	M1	Attempt at the quotient rule. Numerator must be the **difference** of two terms (either way round) of the form $k\cos 3x(2x-4)^3 - l\sin 3x(2x-4)^2$, $k = \pm 3$ $l \geqslant 3$ OR accept: (terms wrong way around) $l\sin 3x(2x-4)^2 - k\cos 3x(2x-4)^3$, $k = \pm 3$ $l \geqslant 3$ Denominator must be of the form $(2x-4)^6$ or accept $\left[(2x-4)^3\right]^2$
	A1	Either term correct, either way round.
	A1	Fully correct
	ALT	
	M1	Use of product rule to give an expression of the form $k\cos 3x(2x-4)^{-3} - l\sin 3x(2x-4)^{-4}$, $k = \pm 3$ $l \geqslant 3$ Accept for this mark $l\sin 3x(2x-4)^{-4} - k\cos 3x(2x-4)^{-3}$
	A1	Either term correct.
	A1	Fully correct
	NB: Ignore any subsequent simplification of their derivative.	

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 9 of 10 · Question 5 part a · 5 marks

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Question number	Scheme	Marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 10 of 10 · Question 9 part a · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

9 (a)	$f'(x) = 4 \times 2x(1+x^2)^3 = 8x(1+x^2)^3$ *	M1 A1* cso [2]
(b)	Find the gradient of the tangent to C at A:	
	$\frac{dy}{dx} = \frac{3\cos 3x(1+x^2)^4 - 8x(1+x^2)^3 \sin 3x}{(1+x^2)^8}$	M1A1A1
	$= \frac{(1+x^2)^3[3(1+x^2)\cos 3x - 8x\sin 3x]}{(1+x^2)^8}$	
	$= \frac{3\cos 3x + 3x^2\cos 3x - 8x\sin 3x}{(1+x^2)^5}$	
	$\frac{3\cos\left(3\times\frac{2\pi}{3}\right) + 3\left(\frac{2\pi}{3}\right)^2\cos\left(3\times\frac{2\pi}{3}\right) - 8\times\frac{2\pi}{3}\sin\left(3\times\frac{2\pi}{3}\right)}{\left(1+\left(\frac{2\pi}{3}\right)^2\right)^5}$	M1
	$= \frac{3\left(1+\left(\frac{2\pi}{3}\right)^2\right)}{\left(1+\left(\frac{2\pi}{3}\right)^2\right)^5} = \frac{3}{\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4}$	A1
	Find the negative reciprocal:	
	$= -\frac{1}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4$	M1A1* cso [7]
ALT (b)	Find the gradient of the tangent to C at A:	
	Using product rule from $y = \sin 3x\left(1+x^2\right)^{-4}$	M1
	$\frac{dy}{dx} = \sin 3x\left(-8x\left(1+x^2\right)^{-5}\right) + 3\cos 3x\left(1+x^2\right)^{-4}$	A1A1
	$\frac{dy}{dx}\left(\frac{2\pi}{3}\right) = \sin\left(3\times\frac{2\pi}{3}\right)\left(-8\times\frac{2\pi}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^{-5}\right) + 3\times\cos\left(3\times\frac{2\pi}{3}\right)\left(1+\left(\frac{2\pi}{3}\right)^2\right)^{-4}$	M1
	$= \frac{3}{\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4}$ oe	A1
	Find the negative reciprocal:	
	$= -\frac{1}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4$	M1A1* cso [7]
		Total 9 marks

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 10 of 10 · Question 9 part a · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
9 (a)	M1	For showing an intermediate equation in which are shown multiplication by the original index, multiplication by the derivative of the bracketed term and a reduction in the index of the bracketed term. Should be in the form $ax(1+x^2)^3$
	A1*cso	For a fully correct solution, minimum steps shown, no omissions or error. Accept "$\frac{dy}{dx}=$" or "$\frac{d}{dx}=$"
(b)	M1	For a minimally acceptable attempt to differentiate using the quotient rule • $\sin 3x \Rightarrow k\cos 3x \quad k \neq 0,1$ where k is an integer • $(1+x^2)^4$ as given in (a) • Allow + /- in the numerator • Denominator must be correct
	A1	For both terms in the numerator, in either order
	A1	For the correct expression shown oe **NB: This mark appears as an M mark in ePen**
	M1	For substitution of $x=\frac{2\pi}{3}$ into any changed expression representing the gradient of the tangent. **NB: This mark appears as an A mark in ePen**
	A1	For the correct expression shown. This may be implied by further correct work. **NB: This mark appears as an M mark in ePen**
	M1	For attempting to find the negative reciprocal of their expression.
	A1cso	For the correct expression shown
ALT (b)	M1	For a minimally acceptable attempt to differentiate using the product rule • $\sin 3x \Rightarrow k\cos 3x \quad k \neq 0,1$ where k is an integer • $(1+x^2)^4$ as given in (a) • Allow + /- between the terms
	A1	For one term correct
	A1	For the correct expression shown oe **NB: This mark appears as an M mark in ePen**
	M1	For substitution of $x=\frac{2\pi}{3}$ into any changed expression representing the gradient of the tangent. **NB: This mark appears as an A mark in ePen**
	A1	For the correct expression shown This may be implied by further correct work **NB: This mark appears as an M mark in ePen**
	M1	For attempting to find the negative reciprocal of their expression.
	A1cso	For the correct expression shown

Type MS · Product rule and chain rule: differentiating products and 'function of a function'

Answer 10 of 10 · Question 9 part a · 2 marks

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Question number	Scheme	Marks